ECON 3200 Discussion Note - Sep 9, 2026
Review: Returns to Scale and Marginal Product

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1 What are returns to scale?

Suppose output is produced according to

$$Q = F(L, K),$$

where L is labor and K is capital.

Returns to scale asks:

> What happens to output if we increase **all** inputs by the same proportion?

For example, suppose we double both labor and capital:

$$(L, K) \to (2L, 2K).$$

We compare

$$F(2L, 2K) \quad with \quad 2F(L, K).$$

More generally, if all inputs are multiplied by some factor $\lambda > 1$, compare

$$F(\lambda L, \lambda K) \quad with \quad \lambda F(L, K).$$

2 Constant, increasing, and decreasing returns to scale

Constant returns to scale

A production function has **constant returns to scale (CRS)** if

$$\boxed{F(\lambda L, \lambda K) = \lambda F(L, K)}$$

for any $\lambda > 0$.

If all inputs double, output doubles. If all inputs triple, output triples.

Increasing returns to scale

A production function has **increasing returns to scale (IRS)** if

$$\boxed{F(\lambda L, \lambda K) > \lambda F(L, K)}$$

for $\lambda > 1$.

If all inputs double, output more than doubles. Possible sources of increasing returns include specialization, fixed costs, or technologies that become more productive at a larger scale.

Decreasing returns to scale

A production function has **decreasing returns to scale (DRS)** if

$$\boxed{F(\lambda L, \lambda K) < \lambda F(L, K)}$$

for $\lambda > 1$.

If all inputs double, output less than doubles. Possible reasons include coordination or management difficulties as the scale of production becomes larger.

3 One-input production: Ricardian model

Suppose labor is the only input:

$$Q = F(L).$$

Returns to scale now asks what happens to output when labor is scaled up.

In the Ricardian model, production is

$$Q = F(L) = MPL \times L,$$

where MPL is **constant**.

If we multiply L by λ,

$$Q' = MPL \times \lambda L = \lambda Q.$$

Therefore,

$$\boxed{\text{With one input, a constant marginal product implies constant returns to scale.}}$$

For a one-input technology with zero output when labor is zero, the following statements are equivalent:

$$\boxed{\text{CRS} \iff \text{linear production function} \iff \text{constant MPL.}}$$

4 Two-input production: Specific-factors model

The connection between CRS and constant marginal product is special to the one-input case.

Suppose output depends on labor and capital:

$$Q = F(L, K).$$

Constant returns to scale requires

$$F(\lambda L, \lambda K) = \lambda F(L, K).$$

The key point is that **all inputs must be scaled together**.

Specific-Factor Model: Cobb–Douglas production

Consider

$$Q = L^{1/2} K^{1/2}.$$

If both inputs double,

$$Q' = (2L)^{1/2} (2K)^{1/2}.$$

Then

$$Q' = 2L^{1/2} K^{1/2} = 2Q.$$

Therefore, this production function has constant returns to scale.

But the marginal product of labor is not constant

The marginal product of labor is

$$MPL = \frac{\partial Q}{\partial L} = \frac{1}{2} L^{-1/2} K^{1/2}.$$

Holding K fixed, MPL falls as L increases.

Therefore,

$$\boxed{\text{CRS does not generally imply constant marginal products.}}$$

5 MPL, PPF, and Labor Demand

The shape of the production function also determines the shape of the PPF and the labor demand curve.

Ricardian Model: Linear PPF and Horizontal Labor Demand

In the Ricardian model, marginal products are constant:

$$MPL_C = \text{constant}, \qquad MPL_W = \text{constant}.$$

Because moving one worker from cloth to wheat always has the same effect,

- wheat output rises by MPL_W;
- cloth output falls by MPL_C.

the opportunity cost of wheat in terms of cloth is constant:

$$\frac{MPL_C}{MPL_W}.$$

Using

$$L = L_C + L_W, \qquad L_C = \frac{Q_C}{MPL_C}, \qquad L_W = \frac{Q_W}{MPL_W},$$

we obtain

$$L = \frac{Q_C}{MPL_C} + \frac{Q_W}{MPL_W}.$$

Rearranging,

$$\boxed{Q_C = MPL_C L - \frac{MPL_C}{MPL_W} Q_W.}$$

Therefore, the PPF is a straight line with slope

$$-\frac{MPL_C}{MPL_W}.$$

Under perfect competition, firms hire labor until the wage equals the value of the marginal product of labor:

$$w = P \times MPL.$$

Since MPL is constant in the Ricardian model,

$$\boxed{w = P \times MPL = \text{constant.}}$$

Therefore, the labor demand curve is horizontal.

Specific-Factors Model: Nonlinear PPF and Downward-Sloping Labor Demand

In the specific-factors model, capital is fixed within each sector. Consider the production function

$$Q = L^{1/2}K^{1/2}.$$

Holding K fixed, the marginal product of labor is

$$MPL = \frac{\partial Q}{\partial L} = \frac{1}{2}L^{-1/2}K^{1/2}.$$

As employment increases,

$$\boxed{MPL \text{ falls.}}$$

Therefore, the value of the marginal product of labor also falls:

$$w = P \times MPL = \frac{P}{2}L^{-1/2}K^{1/2}.$$

Hence, the labor demand curve is downward sloping.

Diminishing MPL also changes the shape of the PPF. With two sectors,

$$Q_C = L_C^{1/2}K_C^{1/2}, \qquad Q_W = L_W^{1/2}K_W^{1/2},$$

and

$$L_C + L_W = L.$$

Because the marginal product of labor changes as workers move between sectors, the opportunity cost is no longer constant. Therefore,

$$\boxed{\text{the PPF is nonlinear.}}$$